

SQL Project Report

SUMMARY

This project demonstrates practical SQL skills by analyzing two relational datasets using Microsoft SQL Server Management Studio (SSMS). The objective was to import data, explore database structures, write SQL queries, solve business problems, and apply query optimization techniques.

Objectives

- Understand relationships between tables.
- Perform data exploration.
- Write SQL queries using filtering, grouping, joins, subqueries, and window functions.
- Answer business questions using SQL.
- Improve query performance through indexing.

Dataset

Chinook Database: Contains multiple table on

1. Customers
2. Artists
3. Albums
4. Tracks
5. Invoices
6. Employees

Sample Sales Dataset: Contains sales information including:

1. Orders
2. Products
3. Customers
4. Sales values
5. Countries

Tools Used

- SQL Server Management Studio (SSMS)
- GitHub

Key Findings

- The United States generated the highest revenue.
- Rock music accounted for the largest share of purchases.
- A small group of customers contributed a significant portion of total revenue.
- Revenue increased during the first months of the year(January), indicating seasonal demand.
- Classic cars were accounted to for the highest sales and revenue.
- The Trooper by the artist Iron Maiden sold the highest unit.
- The first month of 2003 generated the highest quarterly revenue.

SQL Skills Demonstrated

- Select
- Where
- Order by
- Group by
- Having
- Aggregate functions
- Inner join
- Left join
- Right join (where applicable)
- Subqueries
- Window functions
- Indexing

Business Questions Answered

- Top-performing customers
- Best-selling products
- Monthly revenue trends
- Customer purchasing behavior
- Country-wise sales
- Average customer spending

Challenges

- Applying common table expressions(CTEs).
- Using window functions.
- Conversion of datatype from date/time2 to Date using Try_convert
- Altering a table successfully